

CRA8-Handover-Developer

Architecture, environments, deployment, and open items

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CRA8 Website — Developer Handover

For the engineer who maintains this after handover. Pair this with:

- `docs/CRA8-Architecture.md` — the content model, routing, publishing rules, the slate grid, undo/redo, how to add a block type or a collection. **Read it.**
- `docs/CRA8-CMS-Handbook.md` — the editor-facing guide (what the client does).
- `docs/CRA8-Creative-Direction-Final.md` — art direction and the founding slate.
- `README.md` and `server/HOSTING.md` — setup notes (see *Known drift* below before trusting every line).

1. Stack

Frontend — React 18 + TypeScript, Vite 5, Tailwind + shadcn/ui (Radix), Framer Motion, React Router 6, TanStack Query 5, react-hook-form + zod. Built to `dist/`. Two HTML entry points: `index.html` (public site) and `admin.html` (admin SPA), split in `vite.config.ts`.

Backend — `server/` : Express 4 (ESM), `pg` (Postgres), `jsonwebtoken` `auth` (bcryptjs hashes), `multer` uploads, `sharp` image processing, `helmet` , `express-rate-limit` , `CORS`. Storage driver is pluggable: `local` | `s3` | `cloudinary` (`server/utils/`).

Database — Postgres. `server/db.js` owns the schema: `CREATE TABLE IF NOT EXISTS` + an append-only `ADD COLUMN IF NOT EXISTS` list that runs on every boot, plus an idempotent `backfill()` . Deploying and restarting *is* the migration. Never edit a line above the bottom of `ADD_COLUMNS` .

2. Repository layout

<code>src/</code>	Frontend
<code>pages/</code>	Public pages (Index, Work, ProjectDetail, Journal, ...)
<code>pages/admin/</code>	Admin screens (AdminProjects, AdminPages, AdminSettings, ...)
<code>components/site/</code>	Shared render layer – PageShell, Masthead, MediaFrame, SmartImage, Slate, Reveal, Seo, states
<code>components/admin/</code>	FormFields, BlockListEditor, CollectionAdmin, UndoRedo
<code>hooks/</code>	<code>useCmsPage</code> , <code>useContent</code> , <code>useEditHistory</code> , <code>adminQueries</code>
<code>lib/</code>	<code>api.ts</code> , <code>pageBlocks.ts</code> , <code>slate.ts</code> , <code>brand.ts</code>
<code>App.tsx</code>	Routing – explicit routes, then <code>/:slug</code> → <code>SlugPage</code>
<code>server/</code>	
<code>index.js</code>	App wiring; every collection mounted at <code>/api/x</code> and <code>/api/admin/</code>
<code>db.js</code>	Schema + migrations + backfill
<code>routes/</code>	<code>auth</code> , <code>projects</code> , <code>categories</code> , <code>content</code> , <code>journal</code> , <code>services</code> , <code>team</code> , <code>menu</code> , <code>contact</code> , <code>upload</code> , <code>audit</code> , <code>sitemap</code>
<code>utils/collection.js</code>	Shared implementation for <code>journal/services/team</code>
<code>seed.js</code>	Idempotent-ish seed (guarded in prod)
<code>rebrand-cra8.js</code>	DWINDIK → CRA8 data migration (dry-run by default)
<code>docs/</code>	Architecture, CMS handbook, creative direction, this file

Tests: `src/test/cra8-smoke.test.tsx` , `src/test/example.test.ts` (Vitest + Testing Library, jsdom). `npm test` . Coverage is light — a smoke test of the home screen and basics.

3. Local development

Frontend

```
npm install
npm run dev           # Vite on http://localhost:8080, proxies /api → backend
```

Backend

```
cd server
npm install
cp .env.example .env  # then edit – see the env table below
npm run seed          # local admin user + starter content
npm run dev           # nodemon, http://localhost:5001
```

The local `DATABASE_URL` should point at a **local** Postgres

(`postgresql://<user>@localhost:5432/dwindik_cms` on the current dev machine — Homebrew Postgres 16). **Never point localhost at the production database.**

The Supabase URL that may still appear in a local `server/.env` (`db.azaanjaxedjakxlltvxs.supabase.co`) is a **dead, deleted project** and has nothing to do with production. Ignore it / replace it.

4. Environments & deployment

	Frontend	Backend	Database
Host	Vercel	Render web service	Managed Postgres
Name	project <code>dwindik</code> / repo <code>cinematic-dwindik-showcase</code> (<code>finney06s-projects</code>)	<code>dwindik-website</code> (<code>srv-d77kqgc50q8c73csol5g</code>)	via Render <code>DATABASE_URL</code>
URL	the public domain	<code>https://dwindik-website.onrender.com</code>	—
Deploy	auto on push to <code>main</code>	auto on push to <code>main</code> , root dir <code>server/</code> , build <code>npm install</code> , start <code>npm start</code>	schema self-applies on boot
Plan	Hobby (free)	Free instance — sleeps after ~15 min idle, ~50s cold start	free tier = deleted after 90 days

Vercel routing (`vercel.json`): `/admin` and `/admin/*` → `admin.html` ; everything else → `index.html` . To serve the backend sitemap from the frontend domain, add **above** the catch-all:

```
{ "source": "/sitemap.xml", "destination": "https://dwindik-website.onrender.com/sitemap.xml" }
```

Frontend env (Vercel): `VITE_API_BASE_URL=https://dwindik-website.onrender.com/api` .

Render free hours are pooled across the workspace (750 instance-hours/month, shared with `st-agnes-api`). A 24/7 keep-alive pinger alone is ~744 h and will suspend *both* services. The keep-alive is an external **cron-job.org** monitor, scoped to a daytime window (~08:00–16:50 UTC), interval < 15 min. Do **not** re-add the deleted `.github/workflows/keepalive.yml` — two pingers burn hours twice as fast. The real fix for 24/7 uptime is Render **Starter (\$7/mo)**; paid instances don't draw from the free pool.

5. Environment variables (backend)

Var	Purpose	Notes
<code>DATABASE_URL</code>	Postgres connection string	Set in Render dashboard, not in the repo. Managed PG may need <code>?sslmode=require</code> ; <code>server/db.js</code> Pool currently sets no <code>ssl</code> option.
<code>PORT</code>	Listen port	<code>5000</code> on Render, <code>5001</code> local
<code>JWT_SECRET</code>	Token signing	Strong random string in prod
<code>JWT_EXPIRES_IN</code> / <code>JWT_ISSUER</code> / <code>JWT_AUDIENCE</code>	Token config	<code>12h</code> / <code>cra8-cms</code> / <code>cra8-admin</code>
<code>CORS_ORIGIN</code>	Allowed frontend origins	Comma-separated; must include the real frontend domain
<code>SITE_URL</code>	Public origin of the frontend	Used to build <code>/sitemap.xml</code>
<code>ADMIN_USERNAME</code> / <code>ADMIN_PASSWORD</code>	Seed admin defaults	Only used by <code>seed.js</code> ; set a strong password before any prod seed
<code>ALLOW_PROD_SEED</code>	Seed guard	Keep <code>false</code> . Set <code>true</code> only for a deliberate one-off seed, then unset.

Var	Purpose	Notes
<code>MAX_UPLOAD_MB</code>	Upload cap	Default 25
<code>STORAGE_DRIVER</code>	<code>local</code> <code>s3</code> <code>cloudinary</code>	<code>local</code> is not durable on Render — use <code>cloudinary</code> or <code>s3</code> in prod
<code>S3_*</code>	<code>S3_ENDPOINT</code> , <code>S3_BUCKET</code> , <code>S3_REGION</code> (auto for R2), <code>S3_ACCESS_KEY_ID</code> , <code>S3_SECRET_ACCESS_KEY</code> , <code>S3_PUBLIC_BASE_URL</code> , <code>S3_FORCE_PATH_STYLE</code>	if <code>STORAGE_DRIVER=s3</code>
<code>CLOUDINARY_*</code>	<code>CLOUDINARY_CLOUD_NAME</code> , <code>CLOUDINARY_API_KEY</code> , <code>CLOUDINARY_API_SECRET</code> , <code>CLOUDINARY_FOLDER</code>	if <code>STORAGE_DRIVER=cloudinary</code>

`CMS_DB_PATH` appears in `.env.example` and `server/HOSTING.md` — it is a **leftover from the old SQLite build and is unused**. The backend is Postgres-only.

6. Auth & API shape

- Single admin user model (`admin_users`). Login: `POST /api/auth/login` → JWT. Admin requests send `Authorization: Bearer <token>`.
- Every collection router is mounted at both `/api/x` (public, published only) and `/api/admin/x` (token required). `?all=true` / `?draft=true` return unpublished content with a token.
- `lib/api.ts` throws `ApiError` with `status` — pages distinguish 404 (render the 404 page) from backend-down (render a retry). Don't collapse the two.
- `/api/health` has no DB dependency — use it to tell “service down” from “database down”.
- `audit_logs` is written on every mutation, readable at `/api/admin/audit`, and has no UI by design. Keep the writes.

7. Known drift & gotchas

1. **README / HOSTING.md mention SQLite** (`CMS_DB_PATH`, “migrate to Postgres next”). Stale. The app has been Postgres-only since the `chore/postgres-persistence` merge. Worth a docs cleanup pass.

2. **README says the backend is “Express + Postgres”** in one place and implies `npm run seed /Postgres` setup — accurate — but the *frontend* README’s Postgres/S3 sections predate current reality in wording. Verify against `server/HOSTING.md` and this doc.
 3. **`server/db.js` Pool sets no `ssl`**. Some managed Postgres providers reject non-SSL connections or need `?sslmode=require` in the URL. If a fresh DB throws on connect, this is the first thing to check.
 4. **Production DB has gone down once** (Aug 2026) — the free Postgres hit its 90-day deletion. Symptom: `/api/health` 200s, every DB endpoint 500s with `{"error": "Internal server error"}` (each route catch swallows the real error — check Render logs). It has since been restored and is live, but the 90-day clock is the recurring risk. **Fix it permanently:** paid DB plan or scheduled backups.
 5. **Two Render services share one free hour pool**. See §4.
 6. **`rebrand-cra8.js`** exists for DWINDIK→CRA8 data. Dry-run by default; `--apply` to write, `--drop-legacy-projects` also deletes non-slate projects. The frontend already falls back to CRA8 defaults, so this is data hygiene, not a blocker.
 7. **`src/lib/brand.ts`** is the one place identity (name, logo, default menu) is in code, not CMS. Intentional.
 8. **Home (`src/pages/Index.tsx`) is deliberately a single non-scrolling hero** with its own tests — don’t “fix” it into a normal page.
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8. Recovery runbook — production DB down

1. Confirm: `/api/health` → 200, `/api/projects` → 500. It’s the DB, not the app.
 2. Check Render logs for the driver error (ENOTFOUND / auth / SSL).
 3. Provision a new Postgres; set `DATABASE_URL` in Render (add `?sslmode=require` if needed).
 4. Redeploy the backend — `db.js` recreates tables and runs `backfill()` on boot.
 5. Re-seed once: set `ALLOW_PROD_SEED=true` and a strong `ADMIN_PASSWORD` temporarily, run `npm run seed`, then unset `ALLOW_PROD_SEED`.
 6. Re-enter CRA8 content, or restore from a backup if one exists.
 7. Smoke test: `/api/health`, login, `/api/projects`, `/api/menu`, upload a test image and confirm the returned URL is on the storage domain.
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9. Open items / backlog

- ☐ **Make the database durable** — paid plan or automated backups. Highest priority.
- ☐ Decide on Render Starter (\$7/mo) to kill cold starts and pooled-hour risk.
- ☐ Add `ssl` handling to the `pg` Pool config so provider swaps are painless.
- ☐ Clean up `README.md` / `server/HOSTING.md` SQLite references; drop `CMS_DB_PATH`.

- ☐ Log the real error in route catch blocks (or add error middleware detail) so “DB down” is visible without shell access.
 - ☐ Confirm `STORAGE_DRIVER` in production is `cloudinary` or `s3` , not `local` .
 - ☐ Expand test coverage beyond the smoke test.
 - ☐ Run `rebrand-cra8.js --apply` against production if legacy DWINDIK rows remain.
 - ☐ Wire `/sitemap.xml` rewrite in `vercel.json` and set `SITE_URL` on Render.
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10. Credentials to transfer

Hand over, to accounts in the client’s name where possible:

- ☐ GitHub repo access (`Finney06/cinematic-dwin-showcase`)
- ☐ Vercel project (owner/member)
- ☐ Render service `dwindik-website` + workspace note about the shared hour pool
- ☐ `DATABASE_URL` and a fresh database backup
- ☐ Cloudinary / S3 credentials
- ☐ Domain registrar
- ☐ cron-job.org monitor
- ☐ Admin portal username; client sets a new password on first login
- ☐ `JWT_SECRET` (or rotate it and confirm the site still works)